

Red Hat Summit 2026

Third-Party Model Validation

Deploying open-source AI with confidence on Red Hat AI

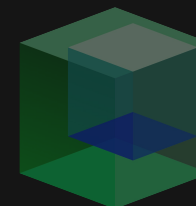
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THE CHALLENGE

Open-source AI is moving fast – but adoption isn't

2.4M+

models on Hugging Face

Growing daily. Innovation is unprecedented – and so is the noise.

6–12 mo

Typical vetting cycle in regulated enterprises

Manual performance, quality and compliance reviews delay every new model from reaching production.

FOUR QUESTIONS BLOCKING ADOPTION



Will it run on my GPU?



Will it be fast enough?



How do I right-size GPU spend?

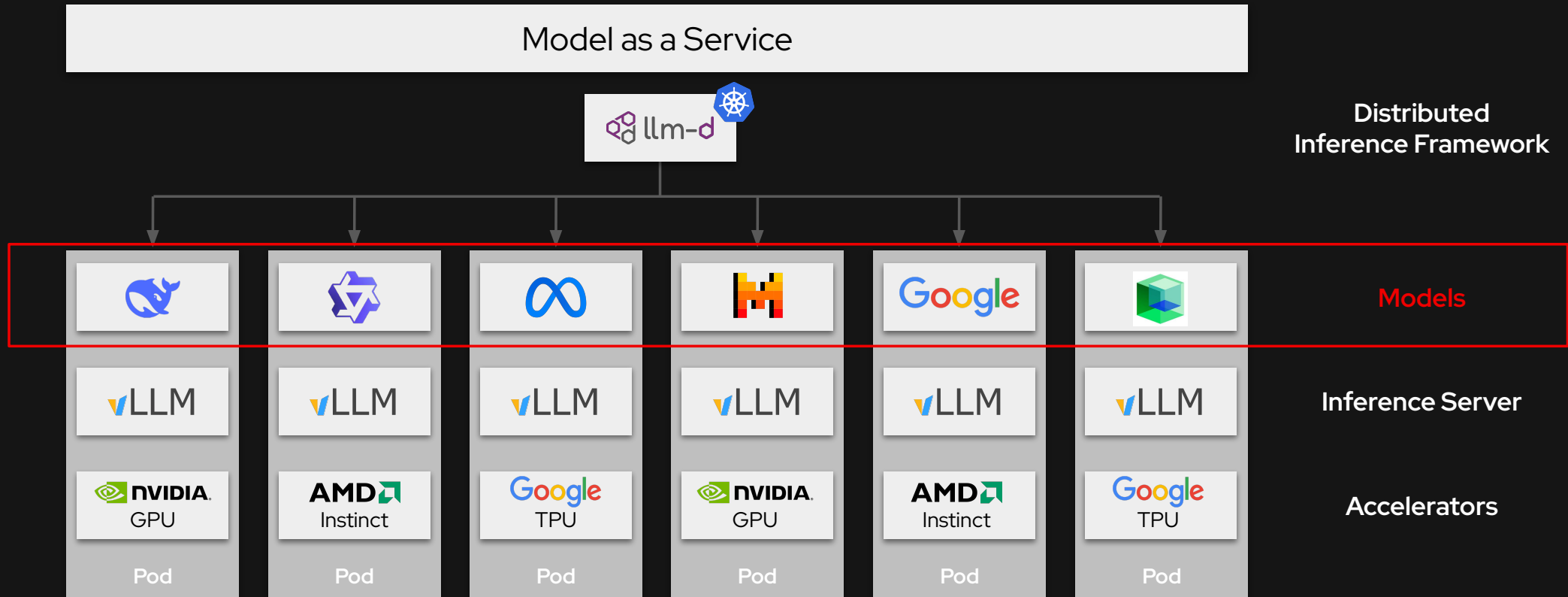


Is the supply chain safe?

Without confident answers, models stay in proof-of-concept and never reach production.

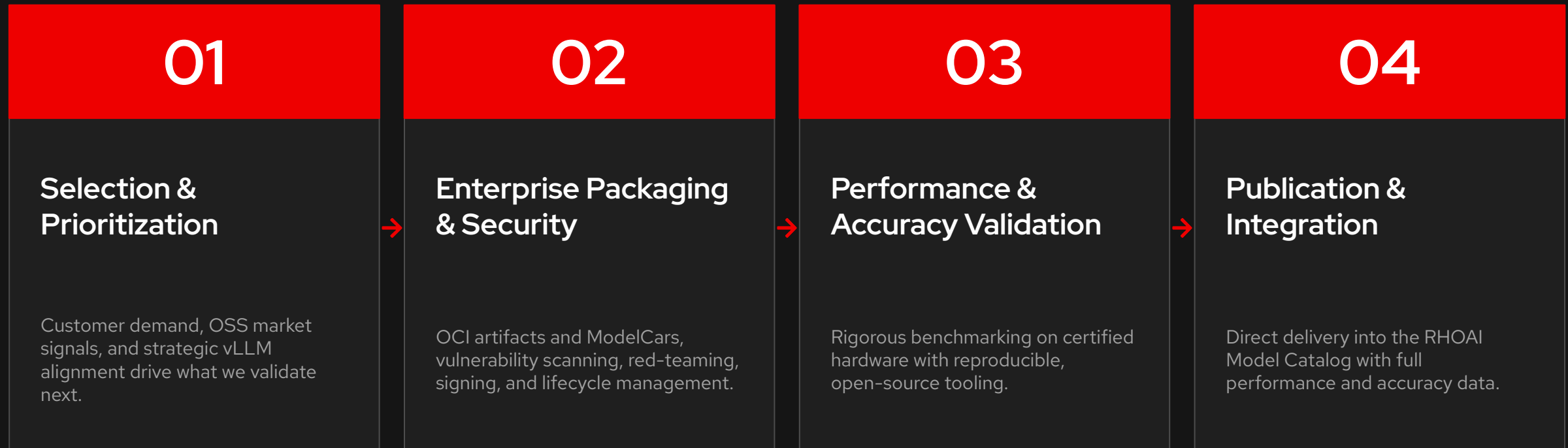
Red Hat AI Enterprise: Enterprise GenAI Inference Platform

Holistic approach to optimize and operationalize deployment and scaling of open-source LLMs



A quantitative framework for validating models

From upstream model to production-ready asset in four stages.



Every stage produces evidence customers can verify, reproduce, and audit.

How we benchmark and orchestrate validation

Open-source tooling. Documented test environments. Reproducible by anyone.



Performance benchmarking

Powered by GuideLLM

- Hardware: NVIDIA A100, H100, B200; AMD Instinct
- Configuration: batch size, max tokens, tensor parallelism
- Workloads: chat, RAG, code, summarization at varied RPS
- Metrics: TTFT, ITL, end-to-end latency, throughput



Accuracy evaluation

Powered by LM-Eval-Harness

- Standardized benchmarks: MMLU, ARC, TruthfulQA, HellaSwag
- Use-case scoring: reasoning, coding, chat, RAG, summarization
- Compressed-vs-baseline: confirms quantization preserves quality
- Folding into TrustyAI for unified evaluation in 1H 2026



Data integrity

Trust through transparency

- Aggregation across runs with statistical summaries
- Outlier detection and monotonicity validation
- Cross-platform validation across NVIDIA + AMD
- Every test, every config, fully documented and reproducible

WHAT CUSTOMERS GET

Delivered directly inside Red Hat OpenShift AI



De-risk adoption

Performance and accuracy data customers need to deploy third-party models without bespoke vetting cycles.



Replace guesswork with data

Hardware-specific profiles and use-case accuracy scores let teams pick the right model and the right GPU footprint.



Secure Models (Roadmap)

Vulnerability scanning, red-teaming, signing, and AI BOMs (roadmap) – the transparency enterprise security teams require.

All modelsRed Hat AI validated modelsRed Hat AI modelsOther models



Validated

[NVIDIA-Nemotron-3-Super-120B-A12B-BF16](#)

Nemotron-3-Super-120B-A12B-BF16 is a large language model (LLM) trained by NVIDIA, designed to deliver...

[View 4 benchmarks](#)

text-to-texttext-generation



Validated

[NVIDIA-Nemotron-3-Super-120B-A12B-FP8](#)

Nemotron-3-Super-120B-A12B-FP8 is a large language model (LLM) trained by NVIDIA, designed to deliver...

[View 3 benchmarks](#)

text-to-texttext-generation



Validated

[NVIDIA-Nemotron-3-Super-120B-A12B-NVFP4](#)

RedHatAI/NVIDIA-Nemotron-3-Super-120B-A12B-NVFP4 is a large language model (LLM) trained by NVIDIA, designed t...

[View 2 benchmarks](#)

text-to-texttext-generation



Validated

[Qwen3.5-397B-A17B-FP8-dynamic](#)

397B Qwen 3.5 MOE model with improved reasoning, tool calling, and VL capabilities, quantized to FP8.

[View 3 benchmarks](#)

image-text-to-text

Red Hat OpenShift AI

Models

CatalogRegistryDeployments

Catalog > NVIDIA-Nemotron-3-Super-120B-A12B-FP8



Validated

NVIDIA-Nemotron-3-Super-120B-A12B-FP8

Provided by NVIDIA

Deploy modelRegister model

OverviewPerformance insights

Hardware Configuration

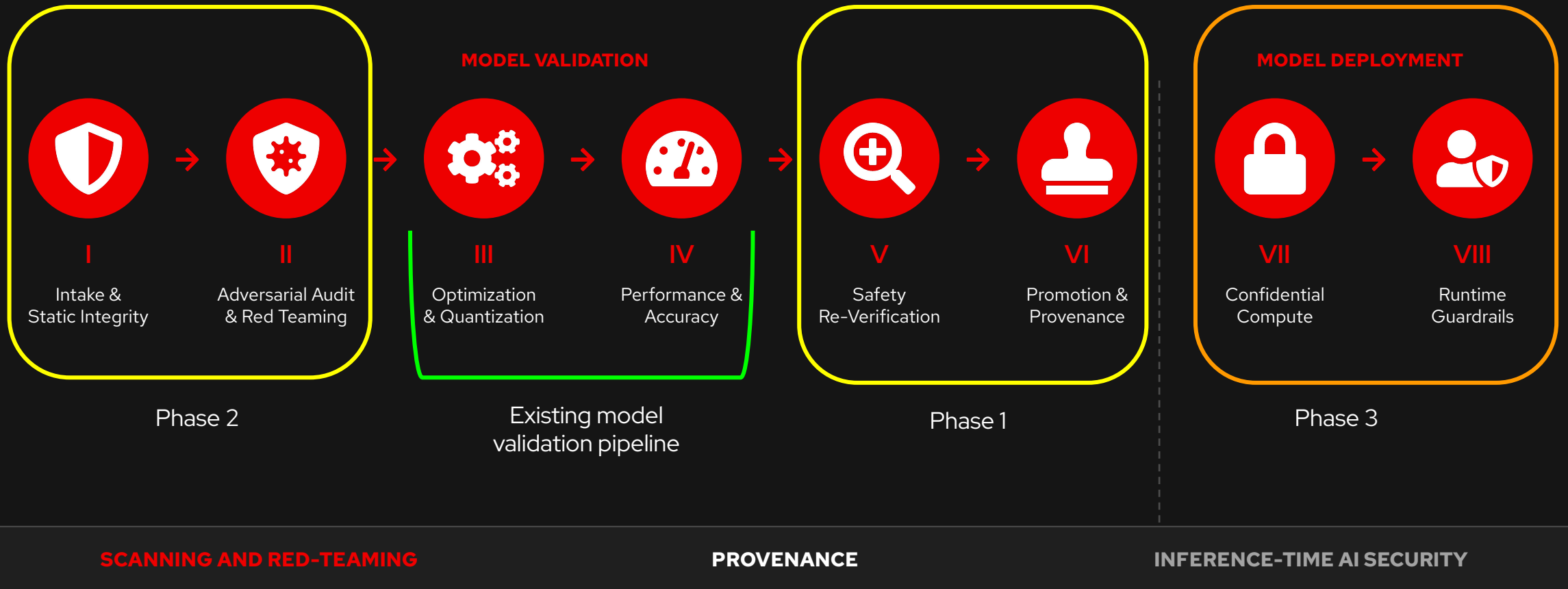
Compare the performance metrics of hardware configuration to determine the most suitable option for deployment.

Scenario: Chatbot (512 input / 256 output tokens)Latency: TTFT at P90 < 2231.15 msMax RPS: 1HardwareCustomize columns

Hardware configuration	Workload type	Replicas	Total RPS	vLLM Version	RPS per Replica	TTFT Latency P90	TPS P90
H200 x 1	Chatbot	1	1	v0.171	1	127.44 ms	370.72 tok/s
H100 x 4	Chatbot	1	2	v0.171	2	136.08 ms	849.91 tok/s
H100 x 2	Chatbot	1	3	v0.171	3	153.54 ms	1372.93 tok/s

The Red Hat AI Model Validation Pipeline Roadmap

Eight stages – from intake through deployment-time guardrails.



Delivering secure models, from the start

Productizing red-teaming and adversarial testing into automated software gates.



Structural integrity

OWASP LLM05

- Pickle-to-SafeTensors conversion
- Static SAST via Snyk and Claire
- Unicode / homoglyph detection
- Malware scanning across all layers



Adversarial resilience

OWASP LLM01

- Garak red-teaming integration
- Prompt injection probes at scale
- Jailbreak and disclosure attacks
- Quantitative AI risk metrics



Runtime guardrails

OWASP LLM01 / LLM09

- TrustyAI bias and safety filtering
- PII and non-public data leak probes
- Confidential compute via TEEs
- Continuous production monitoring

Every validated model passes the same gates – automated, evidence-based, and audit-ready for MRM teams.

Model validation provides guidance on specific scenarios

Task-specific scenarios and reference configurations for the workloads driving enterprise adoption.

TASK-BASED VALIDATION



Tool calling

Q2 2026

Validating function invocation, structured outputs, and chained calls – with files and reproduction commands in the Catalog.



Embedding models

Roadmap

Throughput and quality validation for retrieval workloads – vector quality and downstream RAG performance.



RAG pipelines

Roadmap

End-to-end retrieval, faithfulness, and full-stack latency against reference workloads.

OPTIMIZED vLLM / llm-d CONFIGS



High Throughput

Thank you

Deploy open-source AI on Red Hat AI – with the data, the security, and the confidence to run it in production.

EXPLORE FURTHER

Validated models on Hugging Face

huggingface.co/RedHatAI

Red Hat AI documentation

docs.redhat.com/en/red_hat_ai

Get in touch with your Account Team

via email to learn more

Red Hat is the world's leading provider of enterprise open source software solutions.